

AEGIS-FDIR: Autonomous Embedded Guard & Isolation-Forest Telemetry Anomaly Detector

ESA OSIP Call for Ideas: Autonomous Software Experiments on Hera | Category 5 – Spacecraft Resilience & FDIR



Proposal ID:	ESA-OSIP-HERA-2026-RDAS-FDIR-AEGIS-FDIR	Target Core:	GR712RC LEON3 Core 1 (Bare-Metal @ 50 MHz)
Proposing Entity:	radixal s.r.o. (Brno, Czech Republic)	Memory Limit:	64 kB Stack 18.2 kB Static RAM (0 malloc)
Leadership Triad:	Bc. Viktor Lošák (PI), Ing. Petr Slepíka, Mgr. David Riedl	Standards:	ECSS-E-ST-40C Cat D MISRA-C:2012 Deterministic

EXECUTIVE SUMMARY & KEY IN-FLIGHT BUDGETS:

AEGIS-FDIR is an autonomous onboard telemetry anomaly detection engine that operationalizes the ESA/ESTEC Flight Software Systems Section research (HERA-IoD initiative). Running on Core 1 bare-metal C, it monitors 16 mission telemetry channels via a zero-heap quantized INT8 Isolation Forest.

- **CPU Utilization:** < 1.0% @ 50 MHz SPARC V8 (Peak WCET: 0.12 s per 10-second cycle)
- **RAM Footprint:** 18.2 kB Static RAM | < 8.0 kB Stack (Zero dynamic allocation / No malloc)
- **Monitored Parameters:** 16 continuous Mission Data Pool channels (AOCS, PCDU, SpaceWire, CPS)
- **Anomaly Detection Lead:** Identifies multivariate subsystem degradation 4–12 hours ahead of hard OOL limits
- **Institutional Alignment:** Direct flight demonstration of ESTEC TEC-SW HERA-IoD ADCSS2023 research.

1. Problem Statement & Deep-Space Constraints

Current deep-space health monitoring relies primarily on static Out-Of-Limits (OOL) threshold checks. Static thresholds cannot detect subtle multivariate correlations (e.g. slight temperature rise paired with reaction wheel current drift) that signal component failure long before hard limits are crossed. At Didymos, an anomaly developing during a 20-hour downlink gap may progress to severe fault before Earth operators can intervene.

2. Technical Solution & Algorithmic Architecture

AEGIS-FDIR implements a deterministic quantized INT8 Isolation Forest ensemble:

- **Stage 1 (Telemetry Normalization):** Reads 16 continuous parameters from the Mission Data Pool and performs fixed-point min-max normalization.
- **Stage 2 (INT8 Isolation Forest Ensemble):** 20 micro decision trees stored in static ROM (12.8 kB) compute average path lengths using integer arithmetic.
- **Stage 3 (Fault Isolation & Attribution):** If Anomaly Score > 65%, analyzes tree branch splits to isolate the root subsystem.
- **Stage 4 (PUS Reporting & Events):** Emits routine health scores in PUS Service 3 (SID 0x484) and triggers PUS Service 5 events (ID 0x510) upon critical anomalies.

4. Technical Feasibility & In-Flight Resource Budget

Resource Parameter	Hera Allocation / Limit	AEGIS-FDIR Consumption	Margin
CPU Clock & Execution	50 MHz SPARC V8 (Core 1)	0.12 s WCET per cycle (< 1.0% load)	+99.0% CPU Idle
Stack Depth	64.0 kB max (at 0x40010000)	< 8.0 kB peak stack	+87.5% Stack Margin
Static RAM (BSS+Data)	Bounded Sandbox RAM	18.2 kB Static RAM (0 malloc)	Zero Heap Frag.
Anomaly Lead Time	Ground OOL: 0 hours (Reactive)	4 to 12 hours advance warning	Proactive Recovery
Telemetry Budget	12.0 MB / session budget	128 bytes every 10 min (PUS-3)	< 0.01 MB total

5. Industrial Implementation Roadmap & Leadership Triad

Proposing Entity: Established in 2016 in Brno, Czech Republic, **radixal s.r.o.** specializes in mission-critical embedded systems, safety-critical railway controls (AK Signal / SIL standards), air-gapped defense architectures (URC Systems), national transport backbones (CENDIS / MD ■R), and commercial C-based optical satellite image processing in Norway.

- **Bc. Viktor Lošák (Principal Investigator & Lead Architect):** Over 10 years of embedded systems architecture and mathematical algorithm design. Responsible for overall pipeline design and ESA interface compliance.
- **Ing. Petr Slepíka (Engineering Lead & Delivery Director):** Specialist in safety-critical C software, MISRA-C static verification, automated QEMU CI/CD test harness, and ECSS Cat D qualification.
- **Mgr. David Riedl (Executive Director & Governance):** Responsible for contract governance, institutional compliance with ESA rules, and resource allocation.

6. References & Scientific Baseline

- [1] **López Trescastro, J., et al. (ESA/ESTEC TEC-SW)**, „HERA-IoD: In-Orbit Demonstration of Machine Learning for Anomaly Detection on LEON3 Architectures“, ADCSS2023.
- [2] **Liu, F. T., Ting, K. M., Zhou, Z. H.**, „Isolation Forest“, IEEE ICDM.
- [3] **ECSS Secretariat**, „ECSS-E-ST-40C: Space engineering – Software“, ESA-ESTEC, 2020.